

10 CFR 50.73

RA23-031

June 16, 2023

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2023-001-00, Low Pressure Core Spray
Inoperable due to Minimum Flow Valve Flow Pressure Switch
Failure

In accordance with 10 CFR 50.73(a)(2)(v)(D), Constellation Energy Generation, LLC
(CEG) is submitting Licensee Event Report (LER) Number 2023-001-00 for LaSalle
County Station, Unit 1.

There are no regulatory commitments in this letter. Should you have any questions
concerning this report, please contact Mr. Dan Mearhoff, Regulatory Assurance
Manager, at (815) 415-2800.

Respectfully,



Christopher Smith
Plant Manager
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

LaSalle County Station, Unit 1

☒ 050
☐ 052

2. Docket Number

373

3. Page

1 OF 3

4. Title

Low Pressure Core Spray Inoperable due to Minimum Flow Valve Flow Pressure Switch Failure

5. Event Date

Month	Day	Year
04	17	2023

6. LER Number

Year	Sequential Number	Revision No.
2023	- 01 -	00

7. Report Date

Month	Day	Year
06	16	2023

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100 %

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20		10 CFR Part 50			
☐ 20.2201(b)	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(1)(I)(A)	☐ 50.73(a)(2)(II)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(d)	☐ 20.2203(a)(3)(I)	☐ 50.36(c)(1)(II)(A)	☐ 50.73(a)(2)(II)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2203(a)(1)	☐ 20.2203(a)(3)(II)	☐ 50.36(c)(1)(II)(A)	☐ 50.73(a)(2)(III)	☐ 50.73(a)(2)(IX)(A)	☐ 73.1200(c)
☐ 20.2203(a)(2)(I)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(IV)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(II)	10 CFR Part 21	☐ 50.46(a)(3)(II)	☐ 50.73(a)(2)(V)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(III)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(V)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(IV)		☐ 50.73(a)(2)(I)(A)	☐ 50.73(a)(2)(V)(C)	☐ 73.77(a)(2)(I)	☐ 73.1200(g)
☐ 20.2203(a)(2)(V)		☐ 50.73(a)(2)(I)(B)	☒ 50.73(a)(2)(V)(D)	☐ 73.77(a)(2)(II)	☐ 73.1200(h)
		☐ 50.73(a)(2)(I)(C)	☐ 50.73(a)(2)(VII)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Doug Blair, Acting Operations Director

Phone Number (Include area code)

815-415-2208

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BM	PDS	S382	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

During station surveillance testing, the Low Pressure Core Spray (LPCS) pump discharge flow differential pressure switch malfunctioned, preventing the LPCS pump minimum flow valve from opening on a low flow condition. As a result the LPCS pump was declared inoperable. The cause of the differential pressure switch failure was water leakage into the switch electrical housing caused by a failure of the shaft seal. The faulty switch was replaced and the LPCS pump was restored to operable status.

This condition could have prevented the LPCS system from performing its safety function. This condition is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. There were minimal safety consequences associated with the condition since other emergency safety systems remained operable.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME LaSalle County Station, Unit 1	☒ 050	2. DOCKET NUMBER 373	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 01	REV NO. - 00

NARRATIVE

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The Low Pressure Core Spray (LPCS) system is part of the Division 1 Emergency Core Cooling System (ECCS). LPCS provides low-pressure coolant into the reactor core under accident conditions. The LPCS minimum flow switch, 1E21-N004, monitors pump discharge flow. This flow switch initiates opening and closing the of LPCS minimum flow valve, 1E21-F011.

Condition Prior to Event

Unit(s): 1

Date: April 17, 2023

Time: 0246 CDT

Reactor Mode(s): 1

Mode(s) Name: Power Operation

Power Level: 100%

Description

At 0246 CDT on April 17, 2023, it was discovered that the single train Low Pressure Core Spray (LPCS) system was inoperable. To support Reactor Core Isolation Cooling (RCIC) surveillance testing, the LPCS pump was started for suppression pool mixing. When attempting to secure the LPCS pump, the LPCS minimum flow valve did not open on low flow as expected. It was identified that the pump discharge flow differential pressure switch had failed causing the minimum flow valve to not perform its design function. This resulted in inoperability of the LPCS pump. LPCS was declared inoperable and the appropriate Technical Specification (TS) required actions were entered in accordance with TS 3.5.1, "ECCS - Operating" and TS 3.3.5.1, "Emergency Core Cooling System (ECCS) Instrumentation."

Cause

The cause of the switch failure is process water intrusion into the switch electrical housing. The water intrusion was caused by a failure of the low pressure side u-cup seal around the shaft from the sensing element.

Reportability and Safety Analysis

There were minimal safety consequences associated with the condition since other emergency safety systems remained operable during the period of LPCS inoperability.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(v)(D) as an event or condition that condition that could have prevented fulfillment of a safety function of structures or systems that are needed to mitigate the consequences of an accident. An 8 hour ENS report was made to the NRC at 0937 EDT on April 17, 2023 (EN# 56471).

Safety System Functional Failure Review

The condition is a safety system functional failure (SSFF) defined in accordance with NEI 99-02, Regulatory Assessment Performance Indicator Guidance. The failure of the 1E21-N004 switch caused the minimum flow valve to not open when required.

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2023	- 01	- 00					

NARRATIVE**Corrective Actions**

The failed differential pressure switch was replaced and LPCS was returned to operable status.

Actions created to evaluate extent of condition, parts reliability and design changes.

Previous Occurrences

LER 373-2017-006-00 Low Pressure Core Spray Inoperable due to Minimum Flow Valve Failure in Closed Position

On May 17, 2017, the Unit 1 LPCS minimum flow valve unexpectedly closed when the pump was in standby. The closure was due to a failed SOR differential pressure switch (Model 141 N6-WX7-M9-C1A-JJTTNQ). Water intrusion into the switch electrical housing for the minimum flow switch, 1E21-N004, resulted in a ground fault causing the actuation of logic and subsequent LPCS minimum flow valve closure. The cause was determined to be process water leakage past an internal "U-cup" seal around the shaft from the sensing element into the electrical switch housing.

Component failure data

Manufacturer: Static O-Ring SOR Inc. (S382)

Device (Model): Differential Pressure Switch (141 N6-WX7-M9-C1A-JJTTNQ)

Component ID: LPCS Pump Discharge Flow Bypass Interlock (1E21-N004)